

Koopman Modes & Eigenpairs

①

Consider a discrete-time system:

$$(DS1) \quad X_{k+1} = \underbrace{F(X_k)}_{\mathbb{R}^d \rightarrow \mathbb{R}^d}, \quad k=0,1,2,\dots$$

The Koopman operator acts on observables $g: \mathbb{R}^d \rightarrow \mathbb{C}$,

$$[Kg](x) = [g \circ F](x)$$

Key Idea: Study the linear dynamics of observables,

$$(DS2) \quad g_{k+1} = K g_k, \quad k=0,1,2,\dots$$

Today/ Q1. What can we learn from eigenfunctions of the Koopman operator?

$$\cancel{K\phi = \lambda\phi}, \quad \phi: \mathbb{R}^d \rightarrow \mathbb{C}$$
$$Kv = \lambda v$$

Q2. How can we compute eigenfunctions of the Koopman operator?

Level Sets of Eigenfunctions

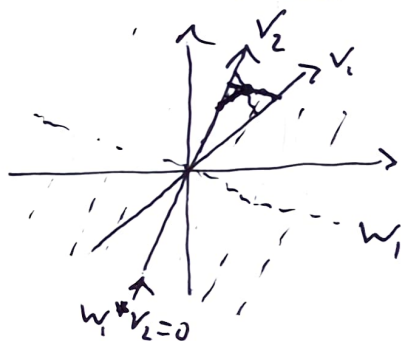
(2) (3)

In linear ^{dynamics} case, we saw that ~~we~~

$\Rightarrow$ zero level sets of Koopman etms correspond to invariant subspaces for the dynamics.

$\Rightarrow$ non-zero level sets of Koopman etms provide notion of "distance" to the invariant subspaces

$$\dot{x} = Ax \Rightarrow w^* A = \lambda w^* \Rightarrow \phi(x) = w^* x$$



$$w_1^* (c_1 v_1 + c_2 v_2) = c_1$$

$$x(t) = c_1 e^{\lambda_1 t} v_1 + c_2 e^{\lambda_2 t} v_2$$

$$\text{if } \lambda_2 > \lambda_1 \Rightarrow x(t) \rightarrow c_2 e^{\lambda_2 t} v_2$$

$$\phi(x(t)) - c_2 e^{\lambda_2 t} v_2^* \phi = c_1 e^{\lambda_1 t} v_1^* \phi$$

$\Rightarrow$ Anything that starts on c_1 -level set, converges to v_2 -invariant set synchronously (cross other level sets at same time!)

In nonlinear case, we have similar properties.

Example. ~~At~~ Zero level sets correspond to invariant sets

if $M_v(0) = \{x \in \mathbb{R}^d \mid v(x) = 0\}$, let $x \in M_v(0)$.

Then,

~~For~~

$$v(F(x)) = [Kv](x) = \lambda v(x) = \lambda \cdot 0 = 0$$

So ~~$F(x) \in M_v$~~

$$x \in M_v(0) \Rightarrow F(x) \in M_v(0)$$

Level Sets of Eigenpairs

Eigenfunctions

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~~Go to Example 1~~

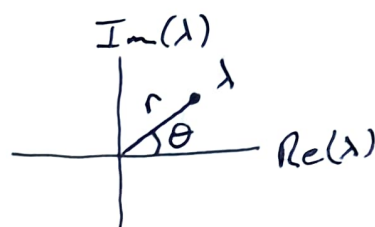
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Suppose that $[Kv](x) = \lambda v(x)$, and write

$v(x)$ and λ in complex polar form,

$$v(x) = R(x)e^{i\theta(x)}, \quad \lambda = re^{i\theta}$$



Let's examine the sublevel sets defined by

"modulus" $M_v(C) = \{x \in \mathbb{R}^d \mid R(x) \leq C\},$

"argument" $A_v(\alpha) = \{x \in \mathbb{R}^d \mid \theta(x) \leq \alpha\}.$

Q: How do these level sets evolve under the dynamics?

$\Rightarrow$ If $x \in M_v(C)$, then ~~$v(F(x))$~~ what can we say about $F(x)$?
or $x \in A_v(\alpha)$

$$v(F(x)) = [Kv](x) = \lambda v(x) = rR e^{i(\theta + \theta(x))}$$

$\Rightarrow$ We can conclude that if $x \in M_v(C)$, then $F(x) \in M_v(rC)$

$\Rightarrow$ Similarly, if $x \in A_v(\alpha)$, then $F(x) \in A_v(\alpha + \theta)$.

Even more precisely,

$$F(M_v(C)) = M_v(rC), \quad F(A_v(\alpha)) = A_v(\alpha + \theta)$$

Level Sets of Eigenfunctions

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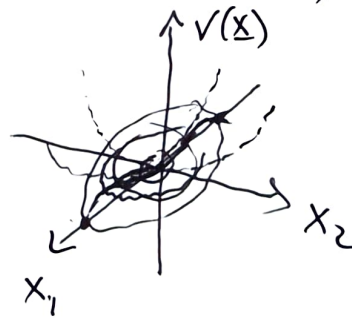
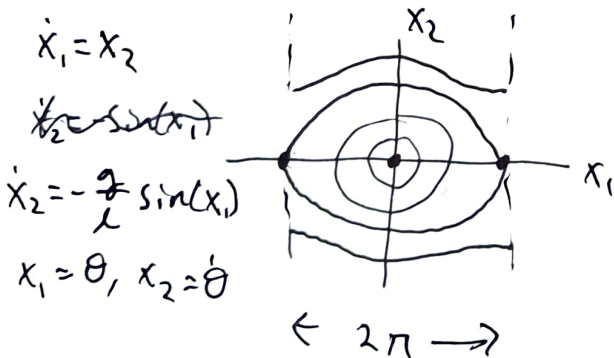
The simple evolution of these sublevel sets allows us to link ~~the~~ key properties of the state space dynamics to the geometry of the Koopman eigenpairs!

Example. If $[Kv](x) = \lambda v(x)$, with $\lambda = 1$, then ~~Not Every level set~~

$$F(M_v(c)) = M_v(c), \quad F(A_v(\alpha)) = A_v(\alpha).$$

$\Rightarrow$ Level sets of ~~invariant~~ eigens w/ $\lambda = 1$ are all invariant sets!

$\Rightarrow$ Systems w/ many periodic orbits often have many eigenfunctions associated w/ $\lambda = 1$. Geometric multiplicity can be high!



If damping is added

Example. If $|\lambda| = 1$, $\omega/\theta = 2\pi p/q \in 2\pi\mathbb{Q}$, then level sets ~~are~~ ^{are} periodic

$$F(M_v(c)) = M_v(c), \text{ and } F(A_v(\alpha)) = A_v(\alpha + \frac{2\pi p}{q})$$

$$\Rightarrow F^q(A_v(\alpha)) = A_v(\alpha + 2\pi p) = A_v(\alpha)$$